

Unit 2 Progress Check: FRQ

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about N_2 and N_2H_4 .

- (a) In the box below, draw the complete Lewis electron-dot diagram of N_2 .



Please respond on separate paper, following directions from your teacher.

- (b) Based on the Lewis electron-dot diagram that you drew, is the N_2 molecule polar? Explain.

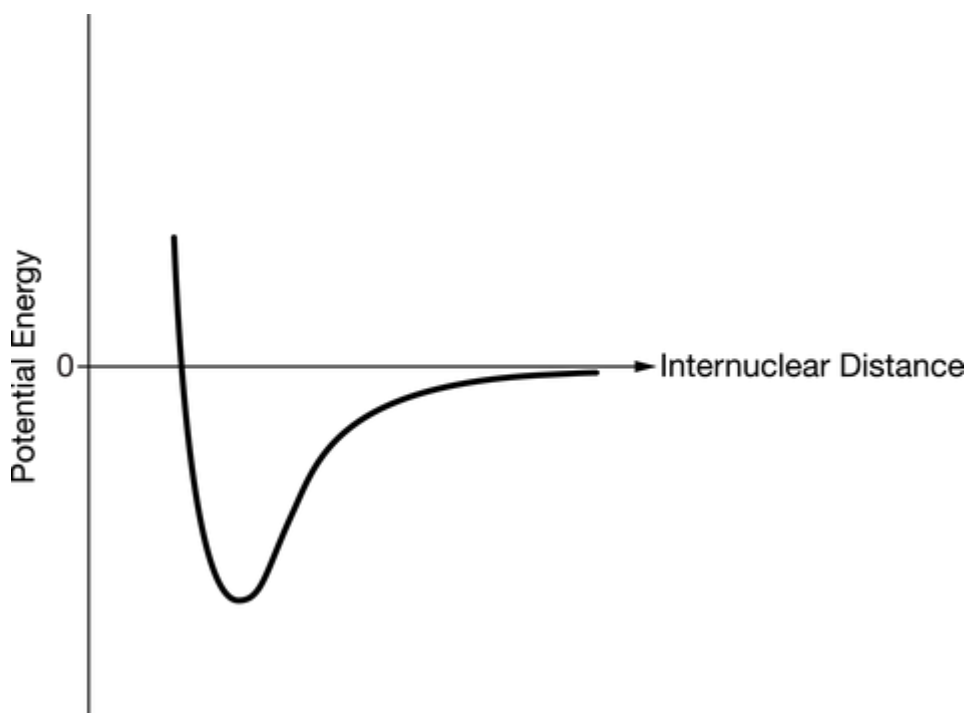


Please respond on separate paper, following directions from your teacher.

The following graph shows the potential energy of two nitrogen atoms versus the distance between their nuclei.



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(c) On the graph, indicate the distance that corresponds to the bond length of the N_2 molecule by placing an **X** on the horizontal axis.



Please respond on separate paper, following directions from your teacher.

(d) On the graph, which shows the potential energy curve of two **N** atoms, carefully sketch a curve that corresponds to the potential energy of two **O** atoms versus the distance between their nuclei.



Please respond on separate paper, following directions from your teacher.

N_2 can react with H_2 to form the compound N_2H_4 .

(e) A sample of N_2H_4 has a mass of **25 g**. Identify the numerical quantity that is needed to convert the number of grams of N_2H_4 to the number of moles of N_2H_4 . (You do not need to do the actual calculation.)

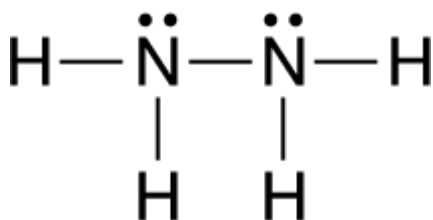


Please respond on separate paper, following directions from your teacher.

(f) The Lewis electron-dot diagram of N_2H_4 is shown below.



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(i) Based on the Lewis electron-dot diagrams of N_2 and N_2H_4 , compare the length of the nitrogen-to-nitrogen bond in N_2 with the length of the nitrogen-to-nitrogen bond in N_2H_4 .



Please respond on separate paper, following directions from your teacher.

(ii) Based on the Lewis electron-dot diagrams of N_2 and N_2H_4 , compare the strength of the nitrogen-to-nitrogen bond in N_2 with the strength of the nitrogen-to-nitrogen bond in N_2H_4 .



Please respond on separate paper, following directions from your teacher.

(iii) Identify the hybridization of the N atoms in N_2H_4 .



Please respond on separate paper, following directions from your teacher.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response meets both of the criteria below:

- ☐ The diagram shows a total of ten electrons.
- ☐ The diagram shows three pairs of electrons between the N two atoms.



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Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the N_2 molecule is not polar, because the three bonding pairs of electrons are shared equally between the two N atoms, and each N atom has a nonbonding electron pair.

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows an **X** drawn on the axis and located directly above the lowest part (minimum) of the potential energy curve. The lowest potential energy on the curve corresponds to the most stable internuclear distance (i.e., bond length).

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The sketched curve meets both of the criteria below:

- ☐ The curve has its minimum to the right of where the given curve has its minimum (because the length of the **O-O** double bond is greater than that of the **N-N** triple bond).
- ☐ The minimum value of the curve is higher than the minimum of the given curve (because the energy of the **O-O** double bond is lower than that of the **N-N** triple bond).



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Part (e)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the molar mass of N_2H_4 (or **32 g/mol**) is needed to solve the problem.

Part (f)(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the nitrogen-to-nitrogen bond in N_2 is shorter than the nitrogen-to-nitrogen bond in N_2H_4 . (No explanation is required, however, the length of a triple bond is shorter than that of a single bond.)

Part (f)(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the nitrogen-to-nitrogen bond in N_2 is stronger than the nitrogen-to-nitrogen bond in N_2H_4 . (No explanation is required, however, the strength of a triple bond is greater than that of a single bond.)

Part (f)(iii)

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response indicates that the hybridization of the **N** atoms is sp^3 . (No explanation is required, however, the **N** atoms in N_2H_4 are surrounded by four electron domains, one of which is a lone pair. This situation corresponds to sp^3 , hybridization.)